On proposals for SMC

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January 8, 2025

1 Feynman-Kac framework for generic SMC

We follow the notations of [Dau and Chopin, 2022]. The sequence of target distributions for $t = 0, \dots, T$, is

$$\pi_t(\mathrm{d}x) = \frac{1}{L_t}\gamma_t(x)\nu(\mathrm{d}x), \tag{1}$$

with $\nu(\mathrm{d}x)$ a base probability measure, γ_t is a measurable non-negative function, $L_t \coloneqq \int \gamma_t(x)\nu(\mathrm{d}x)$ is the normalization constant of π_t. Let $G_t \coloneqq \gamma_t/\gamma_{t-1}$ for $t \geq 1$ and $G_0 = \gamma_0$ be the reweighting functions. Let $M_t(x_{t-1}, \mathrm{d}x_t)$ be a sequence of Markov kernels, s.t. for any t, M_t leaves π_{t-1} invariant.

2 Objective

Let $g : X \times X \to \mathbb{R}$ be some function, and for any $t \geq 0$, let

$$\begin{aligned} E(t) &\coloneqq \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x), Y_{t+1} \sim M_{t+1}(X_t, \mathrm{d}y)}[g(X_t, Y_{t+1})] \\ &= \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)}[\mathbb{E}_{Y_{t+1} \sim M_{t+1}(X_t, \mathrm{d}y)}[g(X_t, Y_{t+1}) \mid X_t]] \end{aligned} \tag{2}$$

Given $E(t)$ monitors the quality of the proposal M_{t+1}, we aim to construct M_{t+1} so to optimize $E(t)$.

3 Adaptive Waste-Free SMC

At iteration $1 \leq t \leq T$, for the particle m and two consecutive particles with same ancestor $1 \leq m \leq M$, $2 \leq p \leq P$,

$$(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}) \sim \pi_{t-1}(\mathrm{d}x_t[p-1]) \otimes M_t(x_t[p-1], \mathrm{d}x_t[p]). \tag{3}$$

Y: why in SMC WF paper, its $\pi_t \otimes M_t$?. Let $t \geq 1$ and $\hat{\pi}_t \coloneqq \sum_{n=1}^N \delta_{X_t^n} W_t^n$, $N = MP$, where $(X_t^n)_{1 \leq n \leq N}$ are the gathered particles $\tilde{X}_t^{m,p}$ for $1 \leq p \leq P$, $1 \leq m \leq M$, then $\hat{\pi}_t$ is an estimate of π_t. Let $(M_{t+1}^\theta)_{\theta \in \Theta}$ be a collection of Markov kernels leaving π_t invariant and let $E(t, \theta)$ be (2) with the dependency in θ made explicit. We aim to construct an estimate of $E(t, \theta)$, let it be $\hat{E}(t, \theta)$, using the samples of iteration t (3) then compute an estimate of $\theta_{t+1}^* = \mathrm{argmax}_{\theta \in \Theta} E(t, \theta)$

$$\hat{\theta}_{t+1}^* = \mathrm{argmax}_{\theta \in \Theta} \hat{E}(t, \theta). \tag{4}$$

We propose, for any $x \in X$, to perform importance sampling from $M_t(x, \mathrm{d}y)$ to $M_{t+1}^\theta(x, \mathrm{d}y)$ (assuming the density ratio $\mathrm{d}M_{t+1}^\theta(x, y)/\mathrm{d}M_t(x, y)$ exists) to estimate the inner expectation $\mathbb{E}_{Y_{t+1} \sim M_{t+1}^\theta(x, \mathrm{d}y)}$, and to estimate the outer expectation $\mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)}$ by the empirical reweighted mean over the particles. For the inner expectation, for all $x \in X$, $\theta \in \Theta$

$$\mathbb{E}_{Y_{t+1} \sim M_{t+1}^\theta(x, \mathrm{d}y)}[g(x, Y_{t+1})] = \mathbb{E}_{Y_t \sim M_t(x, \mathrm{d}y)}[g(x, Y_t) w_{t+1}^\theta(x, Y_t)], \tag{5}$$

with $w_{t+1}^\theta(x, y) \coloneqq \mathrm{d}M_{t+1}^\theta(x, y)/\mathrm{d}M_t(x, y)$. Replacing this expectation by the unbiased estimate obtained with one-sample, and estimating the outer expectation using the reweighted mean over the particles, we obtain the unbiased estimate

$$\hat{E}(t, \theta) \coloneqq \frac{1}{N} \sum_{1 \leq p \leq P, 1 \leq m \leq M} W_t^{m,p-1} g(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}) w_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}). \tag{6}$$

To avoid importance sampling issue, we can consider restrictive objectives or penalised objectives to replace (4), taking into account the effective sample size, the variance of the weights, or replacing Θ by a neighbourhood centered around $\hat{\theta}_t$.

Replication What about computing several $\hat{E}_s(t,\cdot)$, using M' subsets of ancestors and define θ^* has the average of the S argmins $\theta^{s,*}$, or picking $\theta^{s,*}$ that realises the minimum of $\hat{E}_s(t,\cdot)$ [Kleywegt et al., 2002]?

MH kernel At time $t \geq 1$, let $Q_t(x, \mathrm{d}y)$ be some proposal distribution conditionally on $x \in X$. The associated kernel is

$$M_t(x, \mathrm{d}y) = \delta_x(\mathrm{d}y)\left(\int (1 - a_t(x,y))Q_t(x,\mathrm{d}y)\right) + a_t(x,y)Q_t(x,\mathrm{d}y), \tag{7}$$

with $a_t(x,y) = 1 \wedge \frac{\pi_{t-1}(y)Q_t(y,x)}{\pi_{t-1}(x)Q_t(x,y)}$ the Metropolis-Hasting ratio. We use a Rao-Blackwellised estimate for E. Indeed, assume $g(x,x) = 0$, then for any $t \geq 0$, we have

$$E(t) = \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)}\left[\mathbb{E}_{\bar{X}_{t+1}\sim Q_{t+1}(X_t,\mathrm{d}\bar{x})}[a_{t+1}(X_t,\bar{X}_{t+1})g(X_t,\bar{X}_{t+1}) \mid X_t]\right] \tag{8}$$

using proposals $\bar{X}_{t+1} \sim Q_{t+1}(X_t, \mathrm{d}\bar{x})$ before the acceptation-rejection procedure. For any $t \geq 1$, $2 \leq p \leq P$, $1 \leq m \leq M$, let $\bar{X}_t^{m,p} \sim Q_t(\tilde{X}_t^{m,p-1}, \cdot)$ and $\bar{X}_t^{m,1} = X_{t-1}^{A_t^m}$ where $A_t^m \sim \mathrm{resample}(M, W_{t-1}^{1:N})$ be the proposed particles before acceptation-rejection step at iteration t. Then an estimate of $E(t,\theta)$ is given by

$$\hat{E}_t^\theta = \frac{1}{M(P-1)} \sum_{2\leq p\leq P, 1\leq m\leq M}^{P} \bar{W}_t^{m,p-1}\|\tilde{X}_t^{m,p-1} - \bar{X}_t^{m,p}\|^2 a_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p}) w_{t+1}^{\theta,\prime}(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p}),$$

$$w_{t+1}^{\theta,\prime}(x,y) := \frac{Q_{t+1}^\theta(x,y)}{Q_t(x,y)},\ a_{t+1}^\theta(x,y) := 1 \wedge \frac{\pi_t(y)Q_{t+1}^\theta(y,x)}{\pi_t(x)Q_{t+1}^\theta(x,y)},\ \bar{W}_t^{m,p-1} \propto W_t^{m,p-1}, \sum_{p=2}^{P}\sum_{m=1}^{M} \bar{W}_t^{m,p-1} = 1. \tag{9}$$

While for calibrating M_1^θ, i.e., estimating the optimal θ_1, we estimate (8) using samples from ν

$$\hat{E}_0^\theta = \frac{1}{N}\sum_{n=1}^{N} W_0^n \|X_0^n - \bar{X}_1^n\|^2 a_1^\theta(X_0^n, \bar{X}_1^n),\ \bar{X}_1^n \sim Q_1^{\theta_0}(X_0^n, \mathrm{d}\bar{x}),\ a_1^\theta(x,y) := 1 \wedge \frac{\pi_0(y)Q_1^\theta(y,x)}{\pi_0(x)Q_1^\theta(x,y)}. \tag{10}$$

In the case where $Q(x,y) = Q(y,x)$, the previous function is constant and thus cannot be optimised with respect to θ_1. The estimate (9) is (almost) free to compute since all the involved terms are already computed during the procedure at iteration t, except $Q_{t+1}^\theta(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})$ and $Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$.

See Roberts and Rosenthal [2001] for a review of the optimal scaling of MH kernels.

Random Walk: optimal scaling Proposals of the form $y = x + l\sqrt{n^{-\gamma}}Z$, where $Z \sim \mathcal{N}(0, I_n)$. Assume $\pi = f^{\otimes n}$, Roberts et al. [1997] show that $\gamma = 1$, and the optimal asymptotic scaling l satisfies

$$l = 2.38/\sqrt{I}, I = \mathbb{E}[\log(f)'(X)^2], \tag{11}$$

and yields an acceptance rate of $2\phi(-\sqrt{I}l/2) \approx 0.234$

Experiment Let $\Theta = \mathbb{R}^+$, let $\theta := l$. Let $\pi_0 = \mathcal{N}(0, I_n)$, and let L be such that $L\pi_0 = \mathcal{N}(\mu, \sigma^2 I_n)$. Fix $T \geq 1$, and let $\pi_t = L^{\gamma_t}\pi_0$ with $\gamma_t = t/T$, for $0 \leq t \leq T$. We consider the Adaptative RW proposalY: find ref, at iteration $t \geq 1$,

$$y \sim \mathcal{N}(x, l^2/n\hat{\Sigma}_{t-1}), \tag{12}$$

where $\hat{\Sigma}_{t-1}$ is the estimated covariance matrix of the target distribution π_{t-1} given by

$$\hat{\Sigma}_{t-1}' = (N-1)^{-1}\sum_{1\leq i\leq N} W_{t-1}(X_{t-1}^n - \hat{\mu}_{t-1})^\top(X_{t-1}^n - \hat{\mu}_{t-1}), \hat{\mu}_{t-1} = N^{-1}\sum_{1\leq i\leq N} W_{t-1}^n X_{t-1}^n, \tag{13}$$

The ESJD criterion is obtained with $g(x,y) = \|x-y\|^2$, and the corresponding estimator of (2) given by (9). Since all the intermediary distributions π_t are products of identical unidimensional Gaussian distributions, the maximum of the exact ESJD is achieved for the optimal scaling $l = 2.38$ in (12) when $n \to \infty$. Indeed, let $\pi_t = \mathcal{N}(\mu_t, \sigma_t^2 I_n)$, then a computation shows that $I_t = 1/\sigma_t^2$, thus $y = x + 2.38\sigma_t/\sqrt{n}Z$, with $Z \sim \mathcal{N}(0, I_n)$ is the optimal proposal at iteration $t+1$ targeting π_t. Since $\hat{\Sigma}_t \to \sigma_t^2 I_n$, a.s., as N goes to ∞, the proposal (12) should be close in law to the optimal proposal when $l = 2.38$. We aim to estimate the optimal scaling l obtained via maximising (9). As n goes to ∞ and the number of particles $N = MP$ goes to ∞, the optimal scaling $\hat{l}_t$ should converge to 2.38, for all $1 \leq t \leq T$. More generally, one can replace $\hat{\Sigma}_{t-1}$ in (12) by

$$\hat{\Sigma}_{t-1} = N^{-1}\sum_{1\leq i\leq N} W_{t-1}^i \nabla \log \pi_{t-1}(X_{t-1}^i). \tag{14}$$

Algorithm 1: Adaptative Waste Free SMC with MH proposals

input : Integers $M, P \geq 1$, $N \leftarrow MP$, a FK model (initial distribution ν, functions G_t, parametric MH proposals $(Q_t^\theta)_{\theta\in\Theta}$), a criterion g

for $t \leftarrow 0$ **to** T **do**
  if $t = 0$ **then**
    for $n \leftarrow 1$ **to** N **do**
      $X_0^n \sim \nu(\mathrm{d}x_0)$
      $\bar{X}_1^n \sim Q_1^{\theta_0}(X_0^n, \mathrm{d}\bar{x})$
      $g_1^n \leftarrow \frac{\pi_0(\bar{X}_1^n)}{\pi_0(X_0^n)}$
      $d_1^n \leftarrow g(X_0^n, \bar{X}_1^n)$
  else
    $A_t^{1:M} \sim \mathrm{resample}(M, W_{t-1}^{1:N})$
    for $m \leftarrow 1$ **to** M **do**
      $\tilde{X}_t^{m,1} \leftarrow X_{t-1}^{A_t^m}$
      for $p \leftarrow 2$ **to** P **do**
        $\bar{X}_t^{m,p} \sim Q_t^{\theta_t}(\tilde{X}_t^{m,p-1}, \mathrm{d}x_t)$
        $g_{t+1}^{m,p} \leftarrow \frac{\pi_t(\bar{X}_t^{m,p})}{\pi_t(\tilde{X}_t^{m,p-1})}$
        $d_{t+1}^{m,p} \leftarrow g(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$
        $q_{t+1}^{m,p} \leftarrow Q_t^{\theta_t}(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$
        $U \sim \mathcal{U}[0,1]$
        if $U < 1 \wedge \frac{\pi_{t-1}(\bar{X}_t^{m,p})Q_t^{\theta_t}(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})}{\pi_{t-1}(\tilde{X}_t^{m,p-1})q_{t+1}^{m,p}}$ **then**
          $\tilde{X}_t^{m,p} \leftarrow \bar{X}_t^{m,p}$
        else
          $\tilde{X}_t^{m,p} \leftarrow \tilde{X}_t^{m,p-1}$
  for $n \leftarrow 1$ **to** N **do**
    $w_t^n \leftarrow G_t(X_t^n)$
  for $n \leftarrow 1$ **to** N **do**
    $W_t^n \leftarrow w_t^n / \sum_{n=1}^N w_t^n$
  if $t = 0$ **then**
    $\theta_1 \leftarrow \mathrm{argmax}_{\theta\in\Theta} \sum_{n=1}^N W_0^n d_1^n \times \left(1 \wedge g_1^n \frac{Q_1^\theta(\bar{X}_1^n, X_0^n)}{Q_1^\theta(X_0^n, \bar{X}_1^n)}\right)$
  else
    $\theta_{t+1} \leftarrow \mathrm{argmax}_{\theta\in\Theta} \sum_{2\leq p\leq P, 1\leq m\leq M} \bar{W}_t^{m,p-1} d_{t+1}^{m,p} \left(1 \wedge g_{t+1}^{m,p} \frac{Q_{t+1}^\theta(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})}{Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})}\right) \times \frac{Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})}{q_{t+1}^{m,p}}$

Metropolis Adjusted Langevin Algorithm Proposals of the form $y = x + l^2 n^{-\gamma} \nabla \log \pi(x)/2 + l\sqrt{n^{-\gamma}} Z$, with $Z \sim \mathcal{N}(0, I_n)$. For i.i.d target distributions, Roberts and Rosenthal [1998] show that $\gamma = 1/3$ and the optimal asymptotic scaling l satisfies

$$2\phi(-Jl^3/2) \approx 0.574, J^2 = \frac{1}{48}\mathbb{E}[5\log(f)'''(X)^2 - 3\log(f)''(X)^3]. \tag{15}$$

Experiment Similarly to the previous example, we consider $\pi_t = \mathcal{N}(\mu_t, \sigma_t^2 I_n)$. A computation shows that $J_t^{-2} = 16\sigma_t^6$. Let l' be such $2\phi(-J_t l'^{,3}/2) \approx 0.574$, i.e., $l' \approx 1.04 J_t^{-1/3} = 1.650\sigma_t$. Thus, we consider the following adaptative MALA proposal, at iteration $t \geq 1$

$$y \sim \mathcal{N}(x + l^2 n^{-1/3} \mathrm{diag}\hat{\Sigma}_{t-1} \nabla \log \pi_t(x)/2, l^2 n^{-1/3} \hat{\Sigma}_{t-1}), \tag{16}$$

where $\hat{\Sigma}_{t-1}$ is the empirical covariance estimate of π_{t-1} computed using particles and weights of iteration $t-1$. As N goes to ∞, the optimal scaling $\hat{l}_t$ should be close to ≈ 1.650, for all $1 \leq t \leq T$. Again, instead of approximating expectations under π_{t-1} of quantities involving π_{t-1}, we can directly estimate expectations of quantities inolving $\log \pi_t$ using samples (X_{t-1}^i) and IS. In the general case where $\pi = f^{\otimes n}$, let $\hat{J}_t^2$ be the unbiased estimate of J_t^2 given by

$$\hat{J}_t^2 = 48^{-1} N^{-1} \sum_{1\leq i\leq N} W_{t-1}^i(X_{t-1}^i) \left(5(\nabla^3 \log \pi_t(X_{t-1}^i))^2 - 3(\nabla^2 \log \pi_t(X_{t-1}^i)^\top)^3)\right)^2, \tag{17}$$

where the derivatives are taken component-wise. Let $\hat{l}' := 1.04(\hat{J}_t^2)^{-1/6}$, the proposal is then

$$y \sim \mathcal{N}(x + \hat{l}'^{,2} n^{-3} \nabla \log \pi_t(x)/2, \hat{l}'^{,2} n^{-1/3}). \tag{18}$$

RWMH kernel: Autoregressive proposal Let $C \succ 0$, $\nu(\mathrm{d}x)$ some base probability measure on X, e.g., $\nu(\mathrm{d}x) = \mathcal{N}(0, C)$. Let $\Theta = [-1, 1]$, and let for any $t \geq 1$, $x \in X$, $\rho \in \Theta$, $Q_s^\rho(x, \mathrm{d}y) = \mathcal{N}(\mathrm{d}y \mid \rho x, (1-\rho^2)C)$. At iteration $t \geq 1$, let us denote by $\rho_1, \rho_2, \ldots \rho_t \in \Theta$ the sequence of autoregressive parameters for the proposals $Q_1, Q_2, \ldots, Q_t$.

Taking into account higher-order autocorrelations Let ϕ be some target function for which we are interested in estimating $\pi_0(\phi), \ldots, \pi_T(\phi)$. Let $t \geq 0$, let us choose θ^{t+1} so to maximise the efficiency for the next MH chain with π_t as invariant measure

$$\mathrm{eff}(t, \nu) \coloneqq P^{-1}\mathrm{Var}_{X\sim\pi_t}[\phi(X)]/\mathbb{E}[(\hat{\phi}_t(\nu) - \pi_t[\phi])^2], \tag{19}$$

where

$$\hat{\phi}_t(\nu) \coloneqq \frac{1}{P}\sum_{1\leq p\leq P}\phi(X_{t+1}^{*,p}), \qquad (X_{t+1}^{*,1}, \ldots, X_{t+1}^{*,P}) \sim \pi_t(\mathrm{d}x_{t+1}[1])\prod_{p=2}^{P} M_{t+1}^{\nu}(x_{t+1}^*[p-1], \mathrm{d}x_{t+1}^*[p]). \tag{20}$$

Proposition 3.1.

$$\mathrm{eff}(t, \nu) = \left(1 + \frac{2}{P\rho_t(0)}\sum_{l=1}^{P-1}(P-l)\rho_t(l)\right)^{-1}, \tag{21}$$

where $\rho_t(l) \coloneqq \mathbb{E}[(\phi(X_{t+1}^{*,1}) - \pi_t(\phi))(\phi(X_{t+1}^{*,1+l}) - \pi_t(\phi))]$.

Proof.

$$\begin{aligned}\mathbb{E}[(\hat{\phi}_t(\nu) - \pi_t(\phi))^2] &= P^{-1}\mathrm{Var}_{X\sim\pi_t}(\phi(X)) + 2P^{-2}\sum_{1\leq p<p'\leq P}\mathbb{E}[(\phi(X_{t+1}^{*,p}) - \pi_t(\phi))(\phi(X_{t+1}^{*,p'}) - \pi_t(\phi))] \\ &= P^{-1}\mathrm{Var}_{X\sim\pi_t}(\phi(X)) + 2P^{-2}\sum_{1\leq l\leq P-1}(P-l)\rho_t(l).\end{aligned} \tag{22}$$

Taking the inverse and dividing by $P^{-1}\mathrm{Var}_{X\sim\pi_t}[\phi(X)]$ yields the result. □

Proposition 3.2. *Let* $l = 1$, $\rho_t(1)$ *has unbiased estimate*

$$\begin{aligned}\hat{\rho}_t(1) \coloneqq \sum_{1\leq m\leq M}&\bar{W}_t^{m,p}w_{t+1}^{\nu,'}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1})\phi(\tilde{X}_t^{m,p})\Big\{\phi(\bar{X}_t^{m,p+1})a_{t+1}^{\nu}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1}) \\ &+(1 - a(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1}))\phi(\tilde{X}_t^{m,p})\Big\} - \hat{\pi}_t(\phi)^2,\end{aligned} \tag{23}$$

where $\hat{\pi}_t(\phi) = \sum_{1\leq m\leq M, 1\leq p\leq P} W_t^{m,p}\phi(X_t^{m,p})$.

Proof.

$$\begin{aligned}\mathbb{E}[\phi(X_{t+1}^{*,p})\phi(X_{t+1}^{*,p'})] &= \mathbb{E}_{X_{t+1}^{*,p}\sim\pi_t}[\phi(X_{t+1}^{*,p})\mathbb{E}_{X_{t+1}^{*,p+1}\sim M_{t+1}^{\nu}(X_{t+1}^{*,p}, \mathrm{d}x_{t+1}^{*,p+1})}[\phi(X_{t+1}^{*,p+1}) \mid X_{t+1}^{*,p}]] \\ &= \mathbb{E}_{X_{t+1}^{*,p}\sim\pi_t}[\phi(X_{t+1}^{*,p})\mathbb{E}_{X_{t+1}^{*,p+1}\sim M_{t+1}^{\nu}(X_{t+1}^{*,p}, \mathrm{d}x_{t+1}^{*,p+1})}[\phi(X_{t+1}^{*,p+1})]] \\ &= \mathbb{E}_{X_{t+1}^{*,p}\sim\pi_t}[\phi(X_{t+1}^{*,p})\mathbb{E}_{\bar{X}_{t+1}^{*,p+1}\sim Q_{t+1}^{\nu}(X_{t+1}^{*,p}, \mathrm{d}\bar{x}_{t+1}^{*,p+1})}[\phi(\bar{X}_{t+1}^{*,p+1})a_{t+1}^{\nu}(X_{t+1}^{*,p}, \bar{X}_{t+1}^{*,p+1}) \\ &\quad + (1 - a_{t+1}^{\nu}(X_{t+1}^{*,p}, \bar{X}_{t+1}^{*,p+1}))\phi(X_{t+1}^{*,p})]].\end{aligned} \tag{24}$$

Replacing the outer expectation with weighted sum using $(W_t^{m,p}, X_t^{m,p})$, and using IS function from Q_t to Q_{t+1}^{ν}, denoted by $w_{t+1}^{',\nu}$, we obtain the desired estimate. □

Proposition 3.3. *Let* $l = 2$, *then an unbiased estimate of* $\rho_t(2)$ *is,*

$$\begin{aligned}&\sum_{m,p}\bar{W}_t^{m,p}\phi(\tilde{X}_t^{m,p})w_{t+1}^{\nu,'}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1}) \\ &\times\{a_{t+1}^{\nu}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1})w_{t+1}^{\nu,'}(\bar{X}_t^{m,p+1}, \bar{X}_t^{m,p+2}) \\ &\times(a_{t+1}^{\nu}(\bar{X}_t^{m,p+1}, \bar{X}_t^{m,p+2})\phi(\bar{X}_t^{m,p+2}) + (1 - a_{t+1}^{\nu}(\bar{X}_t^{m,p+1}, \bar{X}_t^{m,p+2}))\phi(\bar{X}_t^{m,p+1})) \\ &+(1 - a_{t+1}^{\nu}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+1}))w_{t+1}^{\nu,'}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+2}) \\ &\times(a_{t+1}^{\nu}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+2})\phi(\bar{X}_t^{m,p+2}) + (1 - a_{t+1}^{\nu}(\tilde{X}_t^{m,p}, \bar{X}_t^{m,p+2}))\phi(\tilde{X}_t^{m,p}))\} \\ &-\hat{\pi}_t(\phi)^2\end{aligned} \tag{25}$$

Proof.

$$\mathbb{E}[\phi(X_{t+1}^{*,p})\phi(X_{t+1}^{*,p+2})] = \mathbb{E}_{X_{t+1}^{*,p}\sim\pi_t}[\phi(X_{t+1}^{*,p}) \times \mathbb{E}_{X_{t+1}^{*,p+1}\sim M_{t+1}(X_{t+1}^{*,p},\mathrm{d}x_{t+1}^{*,p+1})}[\mathbb{E}_{X_{t+1}^{*,p+2}\sim M_{t+1}(X_{t+1}^{*,p+1},\mathrm{d}x_{t+1}^{*,p+2})}[\phi(X_{t+1}^{*,p+2}) \mid X_{t+1}^{*,p+1}, X_{t+1}^{*,p}] \mid X_{t+1}^{*,p}]]. \tag{26}$$

The second expectation is equal to

$$\begin{aligned}&\mathbb{E}_{\bar{X}_{t+1}^{*,p+1}\sim Q_{t+1}^{\nu}(X_{t+1}^{*,p},\mathrm{d}\bar{x}_{t+1}^{*,p+1})}[\mathbb{E}_{X_{t+1}^{*,p+2}\sim M_{t+1}(\bar{X}_{t+1}^{*,p+1},\mathrm{d}x_{t+1}^{*,p+2})}[\phi(X_{t+1}^{*,p+2}) \mid \bar{X}_{t+1}^{*,p+1}]a_{t+1}^{\nu}(X_{t+1}^{*,p},\bar{X}_{t+1}^{*,p+1})] \\ &+ (1-a_{t+1}^{\nu}(X_{t+1}^{*,p},\bar{X}_{t+1}^{*,p+1}))\mathbb{E}_{X_{t+1}^{*,p+2}\sim M_{t+1}(X_{t+1}^{*,p},\mathrm{d}x_{t+1}^{*,p+2})}[\phi(X_{t+1}^{*,p+2}) \mid \bar{X}_{t+1}^{*,p}]],\end{aligned} \tag{27}$$

the first inner expectation in (27) is estimated by

$$w_{t+1}(\bar{X}_t^{*,p+1},\bar{X}_t^{*,p+2}) \times (a_{t+1}(\bar{X}_t^{*,p+1},\bar{X}_t^{*,p+2})\phi(\bar{X}_t^{*,p+2}) + (1-a_{t+1}(\bar{X}_t^{*,p+1},\bar{X}_t^{*,p+2}))\phi(\bar{X}_t^{*,p+1})), \tag{28}$$

the second can be estimated similarly but replacing all the proposed particles $\bar{X}_t^{*,p+1}$ by $\tilde{X}_t^{*,p}$. The outer expectation in (27) can be estimated using IS weights, and finally the outer expectation in (26) is estimated using $(W_t^{m,p}, X_t^{m,p})$. □

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